
Explicit Null Congruence Evolution and Redshift Accumulation in Mass-Deficit Node Basin Geometry

James J. Coit

Abstract

We present an explicit derivation of null congruence evolution and associated redshift accumulation within expansion-dominated regions organized by curvature-minimum spacetime structures, previously identified as Mass Deficit Nodes (MDNs). Starting from the Szekeres Class I metric, we derive the effective interior geometry of an MDN basin without imposing symmetry assumptions, showing that a locally isotropic expansion regime arises naturally as a consequence of curvature minimization and suppressed anisotropy. Using the null Raychaudhuri equation, we obtain a closed-form solution for the expansion scalar along null geodesics under conditions of reduced focusing. Integration over affine parameter yields a logarithmic expression for the accumulated redshift, demonstrating that the previously proposed linear scaling law emerges as a leading-order approximation of a fully relativistic result. The derivation establishes a direct connection between spacetime geometry, congruence dynamics, and observable redshift contributions, providing a concrete realization of the MDN optical mechanism within general relativity.

1. Introduction

In earlier work, we introduced Mass Deficit Nodes (MDNs) as curvature-minimum spacetime structures that organize extended basin regions characterized by suppressed matter density and expansion-dominated dynamics. It was shown that null congruences traversing such regions experience persistent divergence due to suppression of the null focusing functional

$$\mathcal{F} = R_{ab}k^ak^b + \sigma^2,$$

leading to a cumulative, path-integrated contribution to redshift.

While this mechanism was established at a conceptual and optical level, its explicit realization within a concrete spacetime framework remains necessary to demonstrate that the effect arises directly from the dynamics of general relativity.

In this work, we derive null congruence evolution explicitly within an MDN basin. Starting from the Szekeres Class I metric, we show that the basin interior naturally reduces to an effective isotropic expansion region. We then solve the null Raychaudhuri equation under suppressed

focusing conditions and compute the resulting redshift accumulation. This provides the first explicit demonstration that the MDN optical mechanism follows directly from GR without additional assumptions.

2. Interior Geometry of MDN Basins from the Szekeres Metric

2.1 Szekeres Class I Metric

We begin with the Szekeres Class I line element:

$$ds^2 = -dt^2 + \frac{(R' - RE'/E)^2}{1 - k(r)} dr^2 + \frac{R^2}{E^2} (dp^2 + dq^2),$$

where:

- $R(t, r)$ is the areal radius,
- $k(r)$ is the curvature function,
- $E(r, p, q)$ encodes anisotropic structure.

This metric provides a general inhomogeneous solution without imposed symmetries.

2.2 MDN Basin Conditions

An MDN basin is structured around a curvature minimum, implying:

1. Curvature regularity near the node:

$$k(r) = k_0 + \mathcal{O}(r^2), |k_0| \ll 1$$

1. Density suppression:

$$\rho \rightarrow 0$$

1. Anisotropy suppression:

$$\frac{E'}{E} \rightarrow 0, \nabla_{p,q} E \rightarrow 0$$

These conditions are not imposed arbitrarily—they arise from the geometric organization of the basin away from ridge and filament boundaries.

2.3 Radial Metric Reduction

The radial component is

$$g_{rr} = \frac{(R' - RE'/E)^2}{1 - k(r)}.$$

In the basin interior:

- $E'/E \rightarrow 0$,
- $k(r) \rightarrow k_0 \ll 1$,

so:

$$g_{rr} \approx \frac{(R')^2}{1 - k_0} \approx (R')^2.$$

2.4 Angular Structure

The transverse component is

$$g_{\perp} = \frac{R^2}{E^2} (dp^2 + dq^2).$$

With $E \rightarrow \text{constant}$:

$$g_{\perp} \approx R^2 d\Omega^2.$$

2.5 Emergent Interior Geometry

The metric reduces to

$$ds^2 \approx -dt^2 + (R')^2 dr^2 + R^2 d\Omega^2.$$

2.6 Regular Expansion Structure

Regularity and smoothness at the curvature minimum imply:

$$R(t, r) = a_{\text{void}}(t) r + \mathcal{O}(r^3).$$

Then:

$$R' = a_{\text{void}}(t).$$

2.7 Resulting Interior Metric

Substitution yields:

$$ds^2 = -dt^2 + a_{\text{void}}^2(t)(dr^2 + r^2 d\Omega^2)$$

2.8 Physical Interpretation

This result shows that:

- an effective isotropic expansion region emerges naturally,
- the expansion rate $a_{\text{void}}(t)$ is local,
- suppression of gradients leads directly to suppression of focusing.

In this limit:

$$R_{ab}k^a k^b \rightarrow 0, \sigma^2 \rightarrow 0, \Rightarrow \mathcal{F} \rightarrow 0.$$

3. Null Geodesics in the Basin Interior

For radial null geodesics:

$$ds^2 = 0 \Rightarrow \frac{dr}{dt} = \frac{1}{a_{\text{void}}(t)}.$$

We define the null tangent vector:

$$k^a = \left(1, \frac{1}{a_{\text{void}}(t)}, 0, 0\right).$$

4. Expansion Scalar and Raychaudhuri Evolution

The expansion scalar is:

$$\theta = \nabla_a k^a = 3 \frac{\dot{a}_{\text{void}}}{a_{\text{void}}}.$$

The Raychaudhuri equation gives:

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \mathcal{F}.$$

In the basin interior:

$$\mathcal{F} \approx 0,$$

so:

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2.$$

5. Solution for the Expansion Scalar

Solving:

$$\theta(\lambda) = \frac{\theta_0}{1 + \frac{1}{2}\theta_0\lambda}.$$

This describes persistent divergence with asymptotic decay.

6. Redshift Accumulation

The redshift contribution is:

$$\Delta z = \frac{1}{3} \int \theta \, d\lambda.$$

Evaluating:

$$\Delta z = \frac{2}{3} \ln \left(1 + \frac{1}{2} \theta_0 L \right)$$

7. Scaling Limit

For moderate $\theta_0 L$:

$$\Delta z \approx \frac{1}{3} \theta_0 L.$$

This reproduces the previously derived scaling law.

8. Observable Formulation

Define:

- L : basin path length
- D : total distance
- $f = L/D$
- $\delta = \theta_0/H - 1$

Then:

$$\Xi_{\text{MDN}} = \frac{\Delta z}{z} \approx \frac{1}{3} f \delta.$$

9. Interpretation

The explicit computation demonstrates that:

- redshift accumulation arises from persistent null divergence,
 - basin interiors dominate due to extended affine length,
 - the effect is non-canceling and path-extensive,
 - logarithmic growth ensures bounded behavior.
-

10. Scope and Limitations

This derivation applies to:

- basin interiors,
- regimes where anisotropy is subdominant.

Not included:

- shear corrections at boundaries,
 - anisotropic $E(r, p, q)$ structure,
 - full null geodesic integration in complete Szekeres geometry.
-

11. Conclusion

We have derived the evolution of null congruences within MDN basin interiors directly from the Szekeres metric and obtained a closed-form expression for the resulting redshift accumulation. This demonstrates that the MDN optical mechanism arises naturally within general relativity and that the previously proposed scaling relation is the leading-order approximation of a fully relativistic result.

This establishes a direct, first-principles connection between curvature-minimum spacetime structure, null congruence dynamics, and observable redshift effects, providing a firm foundation for future studies of anisotropy and observational consequences.